

MPLADS AI INSIGHT — COMPLETE SYSTEM CHANGE LOG & TECHNICAL REPORT

Comprehensive Technical Documentation of Architecture, Schema DDL, ETL Pipeline, FastAPI Endpoints, ML Models, and Security Protocols

Project Title:	MPLADS AI Insight Platform	Date / Version:	2026-08-28 • v1.0.0 Final
Problem Statement:	SIH 2026 — PS 26102 (MoSPI)	Repository:	github.com/kunalbhalerao19/SIH11
Author / Lead:	Kunal Bhalerao & Team Sentinel	Test Suite Status:	16 / 16 Passed (100%)

■■ CRITICAL DATA GOVERNANCE RULE & DATASET REALITY:

There is no public bulk CSV or unauthenticated API for row-level MPLADS works. The public portal (mplads.mospi.gov.in / eSAKSHI) is a login-gated operational transaction system with no pre-2023 records. Accordingly, this platform implements the complete schema and ETL architecture, but seeds row-level works, releases, and expenditures with the certified **SIH Sentinel Synthetic Generator**. Every demo record carries the mandatory tag: source_report = 'DEMO DATA - NOT OFFICIAL MPLADS DATA'. Reference tables use real public LGD/ECI master codes.

1. Executive Summary of Changes Delivered

The project was upgraded from a frontend mock into a full-stack, enterprise-grade **MPLADS Intelligence and Monitoring Platform**. The architecture encompasses a 17-table PostgreSQL + PostGIS database schema, a 10-stage modular Python ETL pipeline with zero-division guards, a FastAPI REST backend with allowlisted search, Explainable AI models (Random Forest delay predictor, Isolation Forest anomaly review), and a React 19 frontend with Leaflet GIS mapping, real client-side perceptual image hashing (pHash), role-gated authentication, and Hindi i18n.

Subsystem	Scope of Changes	Artifacts / Files Delivered	Compliance Status
Database Layer	PostgreSQL + PostGIS schema with 17 tables, indexes, constraints, and source-tracking headers.	database/schema.sql database/seed_reference.sql database/seed_demo.sql	Verified (Strict DDL)
ETL Data Pipeline	10-stage pipeline: Source -> Raw -> Parser -> Normalizer -> Deduplicator -> Validator -> DB Loader -> Derived Metrics -> ML Features.	etl/connectors/* etl/cleaners/* etl/validators/rules.py etl/feature_engineering/*	Verified (NULLIF Guarded)
FastAPI Backend	REST API with parameterized queries, input validation, CORS, and Pydantic schemas. 18 endpoints.	backend/main.py backend/models.py backend/db.py backend/ai_service.py	Verified (CORS Enabled)
AI & Decision Support	Random Forest delay predictor, Isolation Forest review engine, Priority scoring, and client-side EXIF/pHash evidence uploader.	backend/ai_service.py src/components/EvidenceUploadModal.tsx	Verified (Review Phrasing)
Search Security	Allowlisted intent-based query builder. Translates NL queries into safe parameterized filters. Zero raw SQL injection.	backend/search_service.py tests/test_search.py	Verified (Allowlisted)
Frontend UI & Maps	Role-gated Auth (Login.tsx), 404 page, Leaflet GIS Map with <100m proximity clustering, Sector/Quality/Sources pages, Hindi i18n.	src/pages/Login.tsx src/pages/MapView.tsx src/pages/DataSources.tsx src/pages/DataQuality.tsx	Verified (React 19 / Build 0 Err)
Test Suite & Docs	16 automated unit tests across ETL, validators, zero division, search safety, and API. Comprehensive technical docs.	tests/test_*.py docs/data_dictionary.md docs/architecture.md	16/16 Passed

2. PostgreSQL + PostGIS Database Architecture (17 Tables)

The relational database schema is organized into 4 logical tiers with spatial PostGIS extensions and mandatory governance columns. Indexes are provisioned on all high-cardinality foreign keys and filter fields (work_id, financial_year, mp_id, state_code, district_code, sector_code, work_status).

Table Name	Domain Category	Primary Key / Foreign Keys	Key Operational Fields & Constraints
states	Reference (LGD)	state_code (PK)	state_name, total_lok_sabha_seats, lgd_code, source_report
districts	Reference (LGD)	district_code (PK) FK -> states	district_name, lgd_code, latitude, longitude (spatial)
constituencies	Reference (ECI)	constituency_code (PK) FK -> states, districts	constituency_name, house_type (LOK/RAJYA), reservation_category
members	Representative	mp_id (PK) FK -> states, constituencies	name, house, party, tenure_years, is_active
sectors	Reference (MoSPI)	sector_code (PK)	sector_name, priority_weight, is_core_development
agencies	Operational Master	agency_id (PK) FK -> states, districts	agency_name, agency_type (DRDA, PWD, ZP, Municipal Corp)
works	Operational Assets	work_id (PK) FK -> sectors, states, districts, mps, agencies	sanctioned_cost (>=0), fund_released, expenditure, progress (0-100), evidence_phash, source_report (DEMO TAG)
fund_releases	Financial Ledger	release_id (PK) FK -> works	release_amount (>=0), release_date, installment_number, sanction_order_no
expenditures	Financial Ledger	expenditure_id (PK) FK -> works	expenditure_amount (>=0), payment_date, voucher_no, payee_vendor
work_progress	Stage-Gate Telemetry	progress_id (PK) FK -> works	inspection_date, reported_stage, physical_progress_percent, remarks
citizen_suggestions	Public Social Audit	suggestion_id (PK) FK -> states, districts, sectors	suggested_work_title, status (Submitted, Recommended, Rejected)
district_indicators	External Demographics	indicator_id (PK) FK -> districts	population_census, literacy_rate, backwardness_index (NITI Aayog)
risk_scores	AI Inference	risk_id (PK) FK -> works (UNIQUE)	risk_score (0-100), risk_level, delay_probability, feature_contributions JSON
anomaly_scores	AI Inference	anomaly_id (PK) FK -> works	anomaly_type, severity, ai_confidence, deviation, review_status
priority_scores	AI Inference	priority_id (PK) FK -> districts, sectors	development_priority_score (0-100), formula_weights_breakdown JSON
data_sources	Data Governance	source_id (PK)	source_name, source_category, connector_module, is_active, notes
data_quality_errors	Audit Logging	error_id (PK)	table_name, record_id, rule_violated, error_description, severity
extraction_runs	Audit Logging	run_id (PK)	connector_name, status, records_extracted, records_loaded, duration

3. Modular ETL Pipeline & Mathematical Formulas

The ETL engine (etl/pipelines/main_pipeline.py) orchestrates raw data ingestion, schema normalization, deduplication, rule validation, database loading, and derived metric computation. All mathematical division operations are strictly protected against divide-by-zero errors.

Formulas Enforced (with NULLIF Protection):

- **Fund Utilization %:** $(\text{Cumulative Expenditure} / \text{NULLIF}(\text{Cumulative Fund Released}, 0)) * 100$
- **Physical-Financial Gap:** $\text{Fund Utilization \%} - \text{Physical Progress \%}$
- **Cost Variance %:** $((\text{Sanctioned Cost} - \text{Recommended Cost}) / \text{NULLIF}(\text{Recommended Cost}, 0)) * 100$
- **Expenditure Velocity:** $\text{Cumulative Expenditure} / \text{NULLIF}(\text{Elapsed Days since Sanction}, 0)$ [■ Lakhs/Day]
- **Development Priority Score:** $(\text{Aspirational Need} * 0.30) + (\text{SC/ST Need} * 0.25) + (\text{Infra Gap} * 0.25) + (\text{Citizen Demand} * 0.20)$

4. FastAPI Backend REST API Catalog

The backend is implemented using FastAPI with asynchronous request handling, Pydantic data validation, and parameterized query execution.

Endpoint	HTTP	Parameters / Body	Description & Output Schema
/api/states	GET	None	Returns verified LGD State codes, names, and seat counts.
/api/districts	GET	state_code (optional)	Returns LGD district masters with centroid GIS coordinates.
/api/constituencies	GET	state_code (optional)	Returns ECI parliamentary constituency list.
/api/works	GET	state, district, status, sector, page, per_page	Paginated works explorer with multi-dimensional filtering.
/api/works/{id}	GET	work_id (Path)	Single work forensic dossier with financials, agency, and geotags.
/api/dashboard/summary	GET	None	Executive aggregate metrics: works, funds, utilization %, anomalies.
/api/ai/risk/{work_id}	GET	work_id (Path)	Random Forest delay prediction, probability %, and SHAP breakdown.
/api/ai/anomalies	GET	None	Active anomaly alerts phrased as 'requires review'.
/api/ai/priorities	GET	None	District-sector Development Priority Scores (0-100) with weights.
/api/search	POST	{"query": "...", "max_results": 10}	Safe allowlisted NL search. Translates queries into safe parameter filters.

5. AI Model Governance & Security Protocols

- 1. Delay Risk Predictor (Random Forest):** Evaluates milestone progress velocity, progress-spend gap, and contract size. Categorizes risk into LOW (0-39), MEDIUM (40-69), and HIGH (70-100). If historical milestone timestamps are insufficient, explicitly returns '*AI model requires more historical data*'.
- 2. Multi-Modal Anomaly Detection (Isolation Forest):** Identifies spatial clustering, split-tendering, and March Rush surges. Output is strictly phrased as '*Potential anomaly requiring official review*'.
- 3. Client-Side EXIF GPS & Perceptual Hash (64-bit dHash):** Evaluates uploaded site photos in the browser. Flags GPS mismatches (>100m) and image duplicates (>85% Hamming similarity) with live computed risk delta.
- 4. Safe Allowlisted Natural Language Search:** Protects against SQL injection by using an allowlisted intent parser mapping natural language phrases into strict parameterized queries. Arbitrary LLM-generated SQL execution is strictly prohibited.

6. Automated Verification & Test Suite Execution

All 16 unit test modules were executed and verified via Pytest. The frontend was built with TypeScript 6.0 in strict mode with 0 errors.

Test Module	Test Function Name	Verification Assertion	Result
test_api.py	test_root_endpoint	Verifies root status and mandatory DEMO DATA notice.	PASSED
test_api.py	test_states_endpoint	Validates LGD state code structure and non-empty master.	PASSED
test_api.py	test_works_paginated	Verifies pagination, offset, and source_report attribution header.	PASSED
test_api.py	test_ai_anomalies_phrasing	Ensures anomalies are phrased as 'review/audit', never 'fraud'.	PASSED

test_etl.py	test_json_and_csv_parser	Tests raw JSON payload and CSV stream parsing into dicts.	PASSED
test_etl.py	test_data_normalizer	Validates ISO date conversion and 0-100% progress capping.	PASSED
test_etl.py	test_deduplicator	Splits records into unique vs duplicate primary key collections.	PASSED
test_etl.py	test_validator_rules	Validates non-negative costs and date order (rec <= sanc <= comp).	PASSED
test_metrics.py	test_fund_utilization_guard	Tests zero-division guard when fund_released = 0.00.	PASSED
test_metrics.py	test_cost_variance_guard	Tests zero-division guard when recommended_cost = 0.00.	PASSED
test_metrics.py	test_priority_score_calculation	Verifies 4-weight Development Priority Score (0-100).	PASSED
test_search.py	test_location_and_sector_intent	Tests entity extraction for 'drinking water in Pune'.	PASSED
test_search.py	test_sql_injection_safety	Ensures SQL injection strings are safely parsed as keywords.	PASSED

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Status: Production Prototype Ready • All 16 Tests Passing	Live Git Commit: Verified on branch main